DIPLOMA PROGRAMME PRACTICE · SL · SET B

Biology / Paper 1A + 1B

Worked mark scheme · 55 marks · 90 minutes across 1A and 1B

Award one mark per marking point. Credit equivalent reasoning and suitable units. Multiple-choice decisions carry one mark each.

PAPER 1A · QUESTION 1 [1]

Answer C Vaporization requires energy to overcome hydrogen bonding. The energy is drawn from the leaf and its surroundings.

PAPER 1A · QUESTION 2 [1]

Answer D Base pairing gives 3′-TCAGGT-5′. Reading this strand from its 5′ end gives TGGACT.

PAPER 1A · QUESTION 3 [1]

Answer A A nuclear envelope separates eukaryotic chromosomes from cytoplasm. Both prokaryotes and eukaryotes share the other features.

PAPER 1A · QUESTION 4 [1]

Answer B The first name identifies the genus. A binomial alone does not settle detailed relationships or reproductive barriers.

PAPER 1A · QUESTION 5 [1]

Answer A Restricted interbreeding reduces allele exchange. Drift, mutation and selection can then produce divergence.

PAPER 1A · QUESTION 6 [1]

Answer B Equal richness can conceal loss of native species and changes in composition or evenness.

PAPER 1A · QUESTION 7 [1]

Answer C Bends in cis-unsaturated chains reduce efficient packing and weaken intermolecular contacts.

PAPER 1A · QUESTION 8 [1]

Answer C Joining 96 separate residues in one linear chain requires 95 links, each formed in a condensation reaction.

PAPER 1A · QUESTION 9 [1]

Answer C Small nonpolar oxygen molecules dissolve in the hydrophobic core. Ions and large polar molecules face a high barrier.

PAPER 1A · QUESTION 10 [1]

Answer D Compartmentalization permits an acidic lumen and contains hydrolytic enzymes away from most cytosolic substrates.

PAPER 1A · QUESTION 11 [1]

Answer A Microvilli increase membrane area available for absorption while leaving membrane selectivity intact.

PAPER 1A · QUESTION 12 [1]

Answer D Expanding the chamber lowers pressure inside the balloons. Air moves down the resulting pressure gradient.

PAPER 1A · QUESTION 13 [1]

Answer A Removal of phloem blocks sugar translocation across the ring. Intact xylem can still transport water.

PAPER 1A · QUESTION 14 [1]

Answer B With similar shapes, doubling length halves area per volume, reducing the area available for heat loss per unit volume.

PAPER 1A · QUESTION 15 [1]

Answer A Mutualism requires a benefit to both partners. Reproductive measurements with controls provide evidence for both benefits.

PAPER 1A · QUESTION 16 [1]

Answer C Loss that persists after substrate renewal and cooling supports denaturation, whereas substrate exhaustion alone would be reversed.

PAPER 1A · QUESTION 17 [1]

Answer A Both pathways can produce CO_2 in yeast. Oxygen uptake or ethanol production would help distinguish their contributions.

PAPER 1A · QUESTION 18 [1]

Answer B Net carbon dioxide uptake measures photosynthetic fixation less respiratory release. Area and time normalize comparisons.

PAPER 1A · QUESTION 19 [1]

Answer C Vesicles release transmitter on one side and receptors detect it on the other. Diffusion itself is not directional.

PAPER 1A · QUESTION 20 [1]

Answer D Negative feedback opposes the initial disturbance. Restoration of the regulated variable reduces the corrective signal.

PAPER 1A · QUESTION 21 [1]

Answer B Resistance can pre-exist. Antibiotic exposure changes relative survival and reproduction, increasing resistance frequency.

PAPER 1A · QUESTION 22 [1]

Answer B Sampled area is 2.5 m^2. Density is 14 m^{-2} and the extrapolation is 14×20.

PAPER 1A · QUESTION 23 [1]

Answer A A grazer can oxidize organic carbon obtained in food and release it as carbon dioxide.

PAPER 1A · QUESTION 24 [1]

Answer D A parent strand provides the base sequence against which incoming nucleotides pair.

PAPER 1A · QUESTION 25 [1]

Answer B There are 25 triplets. The start codon encodes an amino acid and the terminal stop codon does not.

PAPER 1A · QUESTION 26 [1]

Answer D One added nucleotide changes the reading frame downstream. The earlier sequence is unchanged.

PAPER 1A · QUESTION 27 [1]

Answer B Homologues separate into different products in meiosis I. Sister chromatids separate in meiosis II.

PAPER 1A · QUESTION 28 [1]

Answer C Fertilization restores the diploid chromosome complement. Pollination transfers pollen but does not itself fuse nuclei.

PAPER 1A · QUESTION 29 [1]

Answer A Insulin promotes glucose uptake and storage in responsive tissues. Inadequate secretion slows correction of post-meal hyperglycaemia.

PAPER 1A · QUESTION 30 [1]

Answer D Temperature also responds to other forcings and variability. Two observations do not isolate the long-term effect of CO_2.

PAPER 1B · QUESTION 1 [6]

Comparing habitat surveys

(a) Compare initial species richness and evenness in the two marshes. [2]

1 Both initially have richness of three.

1 X is more even because its abundances are equal whereas J dominates Y.

(b) Calculate the percentage of recorded individuals in marsh Y that belong to J. Explain what the second survey reveals about observed richness. [2]

1 75/90×100=83.3%.

1 Detectability affects measured richness, so the first sample underestimated richness in Y.

(c) Propose a sampling change to compare the marshes more reliably and state how it improves the evidence. [2]

1 Use repeated randomly or systematically located equal-area samples with matched timing and effort.

1 Replicates estimate spatial variation and reduce dependence on a single patch or chance encounter.

PAPER 1B · QUESTION 2 [6]

Permeability of membrane models

(a) Calculate the fraction of channel-associated flux removed by the inhibitor, correcting for the lipid-only background. [2]

1 Without inhibitor the channel-associated flux is 18−2=16 and with inhibitor it is 6−2=4.

1 Fraction removed=(16−4)/16=0.75 or 75%.

(b) Explain why the results support facilitated diffusion and do not establish active transport. [2]

1 An open protein increases flux in the direction of the existing gradient.

1 There is no evidence of uphill movement driven by an energy source.

(c) State why the lipid-only inhibitor control is needed and suggest one further measurement of selectivity. [2]

1 It tests whether inhibitor alters background permeability independent of Z.

1 Measure flux of a second solute through the same membranes under matched gradients.

PAPER 1B · QUESTION 3 [6]

Two exposures to an antigen

(a) Compare antibody concentration seven days after each exposure using a ratio. [2]

1 Compare day 7 with day 35.

1 80/4=20, so the measured concentration seven days after the booster is twenty times as high.

(b) Explain the biological basis for the faster secondary response. [2]

1 Specific memory cells persist after the primary response.

1 They rapidly proliferate and form antibody-secreting cells after re-exposure to the antigen.

(c) Evaluate the claim that these measurements prove the booster prevents disease in all recipients. [2]

1 Antibody concentration is an immune marker and the data come from one person.

1 Protection requires evidence such as infection or disease outcomes in suitable comparison groups across recipients.

PAPER 1B · QUESTION 4 [7]

Counting dividing cells

(a) Calculate the mitotic index in each region. [2]

1 P has 36 mitotic cells out of 180, giving 0.200.

1 Q has 19 out of 190, giving 0.100.

(b) A student says P must have a cell cycle half as long as Q. Evaluate this inference. [2]

1 P has twice the observed fraction in mitosis.

1 The fraction also depends on mitotic duration and on the proportion of actively cycling cells, so it does not alone give cycle length.

(c) Describe a way to test inter-observer reliability without counting different fields. [2]

1 Give both observers the same anonymized images and have them classify independently.

1 Compare their classifications or agreement for individual cells using predefined criteria.

(d) Give a visible criterion distinguishing anaphase from metaphase. [1]

1 In anaphase two sets of chromatids move toward opposite poles instead of being aligned at the equator.